

KANDAHAR AP, Afghanistan

WMO: 409900

Lat: 31.5021N

Lon: 65.8524E

Elev: 3304

StdP: 13.02

Time Zone: 4.50 (AFG)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.5	30.1	-2.3	5.5	44.3	1.3	6.6	44.7	26.6	43.7	24.6	41.5	5.7	60	0.464

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.4	107.9	63.1	106.1	62.7	104.3	62.2	69.1	94.0	66.0	98.3	63.9	101.1	10.3	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.0	84.3	84.0	55.2	73.3	71.4	52.0	65.0	67.1	35.3	94.6	32.6	97.7	30.8	101.3	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.4	18.5	15.9	DB	22.5	112.0	3.7	1.3	19.8	113.0	17.6	113.8	15.5	114.5	12.8	115.5	
			WB	17.7	70.5	3.0	3.2	15.5	72.8	13.8	74.6	12.1	76.4	9.9	78.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	69.6	46.5	50.9	61.7	72.0	81.5	88.3	91.9	88.3	79.0	68.8	57.6	47.7
	DBStd	16.85	6.50	7.88	7.18	6.49	5.05	4.34	3.10	3.93	5.08	5.85	6.53	6.16
	HDD50	355	147	76	4	0	0	0	0	0	0	0	14	114
	HDD65	1935	572	397	150	18	0	0	0	0	1	29	233	535
	CDD50	7515	39	101	366	659	975	1148	1299	1188	870	584	242	44
	CDD65	3622	0	2	48	227	510	698	834	723	421	148	11	0
	CDH74	55574	0	26	601	2832	7147	11278	13993	11384	6014	2076	223	0
	CDH80	34891	0	1	152	1286	4220	7588	9738	7525	3474	870	37	0
Wind	WSAvg	6.6	7.4	8.8	7.0	7.1	7.6	7.3	6.5	6.3	5.8	5.0	4.9	5.2
Precipitation	PrecAvg	7.2	1.2	1.7	1.4	0.8	0.3	0.1	0.2	0.2	0.2	0.2	0.4	0.6
	PrecMax	12.1	2.8	3.7	3.5	3.2	1.7	1.6	1.7	1.0	1.3	1.3	2.4	2.7
	PrecMin	2.8	0.1	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0
	PrecStd	2.2	0.8	0.9	0.8	0.7	0.3	0.3	0.4	0.3	0.3	0.3	0.5	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.2	77.3	88.2	97.1	104.5	110.9	110.0	108.0	101.8	95.4	84.4	71.5
		MCWB	48.7	53.0	56.2	59.8	61.1	63.2	64.0	63.4	60.6	56.9	54.3	48.7
	2%	DB	66.7	73.2	84.2	93.7	102.4	107.7	108.3	105.9	98.9	93.0	80.2	67.9
		MCWB	46.7	50.8	55.3	58.5	61.1	62.7	63.4	63.2	59.3	56.5	52.9	46.8
	5%	DB	63.0	69.0	80.6	91.2	100.3	105.8	107.3	104.0	97.2	89.2	76.0	64.3
		MCWB	45.0	49.9	53.7	57.6	60.5	62.2	63.0	62.5	58.6	55.4	51.0	45.3
	10%	DB	59.2	64.4	76.9	87.7	96.8	103.6	105.5	102.0	95.1	84.5	72.0	60.7
		MCWB	43.4	48.5	52.4	56.3	59.4	61.3	62.7	61.9	57.9	53.7	49.5	44.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.3	56.2	61.2	61.8	63.0	73.6	69.2	71.2	68.0	59.0	58.4	51.7
		MCDB	60.9	66.7	74.5	85.3	96.4	81.6	99.1	94.6	87.3	85.5	68.1	64.4
	2%	WB	49.3	54.3	58.5	60.3	62.1	66.7	66.6	69.7	64.7	57.4	56.0	49.2
		MCDB	60.9	65.4	73.6	83.3	96.1	98.7	101.3	92.8	89.9	86.1	67.4	60.1
	5%	WB	47.5	52.4	56.5	59.3	61.2	63.8	65.0	66.4	60.0	55.9	54.0	47.5
		MCDB	58.9	63.9	72.0	81.8	94.6	102.8	102.3	96.9	96.3	85.7	69.1	58.9
	10%	WB	45.7	50.5	54.8	58.3	60.3	62.2	63.7	63.0	58.2	54.6	52.1	45.9
		MCDB	56.1	61.8	70.2	80.5	92.5	101.7	102.5	100.1	94.4	83.8	68.5	57.2
Mean Daily Temperature Range	5% DB	MDBR	21.8	19.1	23.4	25.2	26.7	27.9	26.4	26.6	30.9	29.7	25.5	23.5
		MCDBR	26.0	24.1	27.4	30.1	31.0	30.2	28.1	27.5	33.1	33.2	29.7	27.1
		MCWBR	13.4	9.7	9.1	10.1	11.4	11.7	11.2	11.3	13.9	14.7	14.2	13.7
	5% WB	MCDBR	20.3	19.2	22.2	24.3	27.2	26.6	25.6	20.7	29.0	30.3	22.7	21.3
		MCWBR	11.7	9.1	8.2	8.2	9.7	10.5	11.3	10.1	12.7	13.4	11.4	11.6
Clear-Sky Solar Irradiance	taub	0.333	0.392	0.437	0.477	0.468	0.459	0.524	0.488	0.388	0.358	0.340	0.317	
	taud	2.264	2.073	1.941	1.870	1.899	1.922	1.777	1.855	2.090	2.169	2.245	2.329	
	Ebn at Noon	279	270	266	260	263	264	247	254	278	278	274	279	
	Edh at Noon	35	47	57	64	63	61	70	64	49	42	35	31	
All-Sky Solar Radiation	RadAvg	1131	1345	1786	2163	2487	2662	2473	2324	2113	1743	1315	1108	
	RadStd	83	124	108	89	77	75	106	105	49	46	62	77	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page